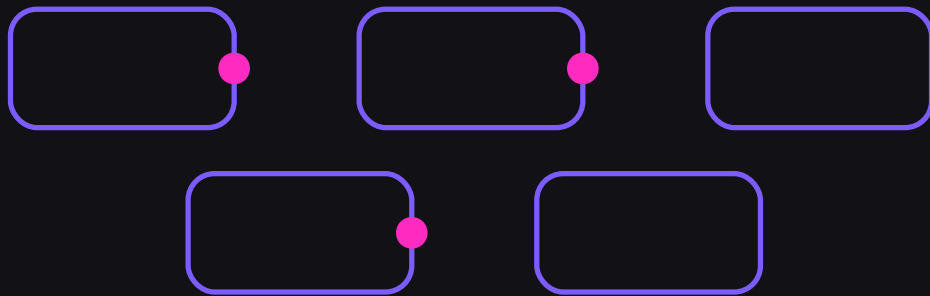


After the Robot

What happens when a machine's body is no longer fixed?

A designed article on robotics, embodiment and the changing boundary between engineered and living systems.



What happens when a machine's body is no longer fixed?

The machine knows what body it has because engineers decided what body it has.

That assumption holds until the first joint wears out.

In 2025, researchers at Columbia University placed robotic arms in front of cameras and let them watch themselves move. The robots had no detailed mathematical description of their own bodies. From short video recordings, they learned models that predicted their morphology, kinematics and movement, good enough to support motion planning.

Then the researchers changed the robots. When the physical body no longer matched the learned simulation, the system detected the mismatch, updated its internal model and recovered useful control.^[1]

Real robots in real environments face this all the time. Joints wear. Payloads change. Actuators weaken. Components fail. Soft structures deform. Modular systems reconfigure. Biohybrid parts develop. A machine that copes needs more than a good model of the outside world. It needs a model of itself.

And that points to a larger shift: the robot may no longer be a finished object.

Engineering usually ends before operation begins

The conventional sequence is familiar: design, manufacture, program, deploy. After deployment the software may change a lot. The body usually doesn't. The legs stay the same length, the sensors stay where they were, the gripper keeps its geometry. The controller can learn new behaviours, but inside a body engineers largely froze in advance.

The earlier articles in this series already complicated that picture. Morphology changes the problem an AI has to solve. Soft and bio-inspired materials distribute sensing and control through the body. Living components keep developing after they are built into a machine.

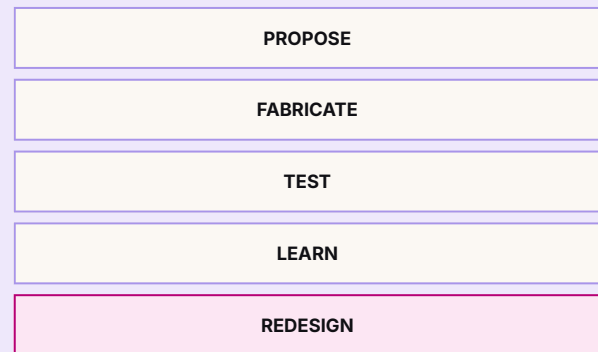
Once the body stays variable, a new question appears: could robot design continue after deployment? Several research directions supply pieces of an answer.

AI enters the design loop

Soft robots make body design hard. Change a chamber shape, wall thickness, stiffness or finger geometry and the behaviour shifts in ways simulation alone predicts poorly.

In 2026, CSIRO researchers introduced FAST (Fast Accurate Soft Testing), an automated platform that combines computational modelling, Bayesian optimisation, 3D printing and physical testing. An optimiser proposes soft-gripper designs, candidates are explored computationally, promising ones are fabricated, robots test them, and the results feed the next round.^[2] Across the reported work, the team ran more than 50,000 simulated grasps and more than 8,500 physical grasps. The combined approach found grippers that outperformed those found by either strategy alone.

DESIGN BECOMES A LOOP



The engineer still sets the objective, the constraints and the design space. The exact geometry comes out of repeated interaction between AI, simulation and the physical world. Once that loop is automated, the line between designing a robot and training it starts to thin.

Deployment doesn't have to freeze learning

The physical world still produces situations no design process anticipated. A motor loses power, a wheel slips, weight shifts, wind pushes. Robots traditionally

handle these through predefined fault-tolerance strategies or controllers built for particular disturbances.

In 2026, researchers introduced FLAIR (Fast Learning-Based Adaptation for Immediate Recovery). While the robot operates, it updates an onboard model every 225 milliseconds, watches how actual motion differs from the requested behaviour, learns the effect of the disturbance and adjusts its commands.^[3]

They tested a tracked robot against altered motor torque, low-friction surfaces, uneven weight distribution and simulated wind. In the tracked-robot experiments, FLAIR recovered 74.4% of the operability of the unperturbed robot and performed substantially better than the comparison approaches in the study.

The robot meets a physical condition it was never trained on and starts adapting while it is happening. Its controller learns the machine it has now, not the one in the original specification.

The controller has to follow the body

That gets harder when the physical change is intentional. Soft robots can be assembled in different lengths, stiffnesses and configurations. Modular robots can change structure entirely. A controller trained on one morphology usually performs badly on another.

In 2026, researchers demonstrated a reinforcement-learning framework across 33 soft-robot configurations, including elephant-trunk-like arms, soft-muscle systems, worm-like robots and combinations. The configurations differed in geometry, stiffness, material properties and assembly. Instead of retraining a controller for each one, the team mapped shared robot dynamics into a common representation and adapted policies there. Transferring control between configurations needed 75 times fewer samples than the study's baselines, including under payload changes and actuator faults.^[4]

As morphology gets less permanent, control has to get more flexible.

A robot can have more than one body

Modular robotics makes the idea literal. In July 2026, an MIT-led team introduced FloatForm, a set of small autonomous robotic boats. Individual modules navigate on their own, connect physically, form larger structures, disconnect and reassemble in other configurations. Connected groups can also travel together across the water. Eight physical modules were demonstrated experimentally, and larger groups were studied in simulation.^[5]

The individual boats persist. The larger machine is temporary. Depending on the task, the same hardware can exist as separate agents, one connected structure, or a different structure later.

That creates an identity problem. Traditionally a robot is tied to one physical shape. In modular systems, identity may sit in the modules, in the controller, in the collective, or in the ability to keep forming useful bodies. The body becomes a configuration, not a definition.

What if the structure knows its own shape?

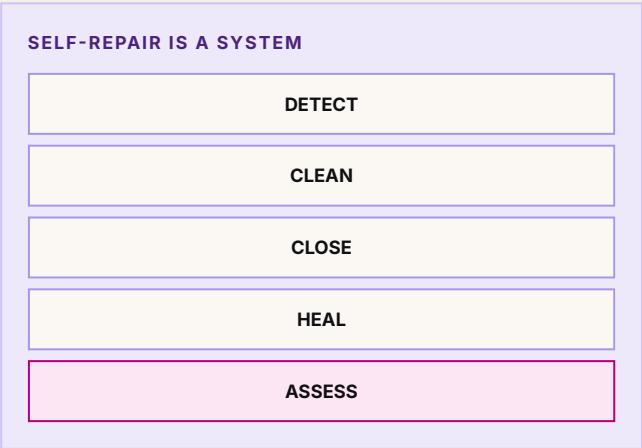
A system of hundreds of interchangeable modules has to understand its current structure. In 2026, researchers tackled this with smart cellular bricks: simple modules with local sensing, computation and communication. Each brick talks only to its neighbours, and nothing sits above them with a complete map.

Using neural cellular automata, collections of hundreds of bricks can collectively infer what type of three-dimensional structure they form. In hardware experiments with almost 200 physical bricks, the researchers demonstrated shape classification and damage detection, locating missing components.^[6] Their computational work also explored more extensive recovery and regeneration, where simulated cellular systems used distributed information to guide restoration of damaged shapes. Those regeneration results remain simulations. The physical experiments showed self-recognition and damage localisation.

The inspiration is biological. No cell holds a blueprint of the whole organism and consults it. Global anatomy emerges and is maintained through huge numbers of local interactions. A future machine may carry its understanding of its own structure in the structure itself.

Damage is several different problems

"Self-healing" makes this field sound further along than it is. Repair is a systems problem. A 2025 review of self-healing soft robotics splits autonomous recovery into five stages: damage detection, cleaning, closure, material healing and recovery assessment. Current systems rarely integrate all five without human help.^[7]



That helps separate capabilities that often get bundled together. A controller can compensate for a weakened motor. A modular machine can reconfigure around a missing part. A material can restore its properties after a cut. A distributed structure can detect where damage happened. Each adds resilience. A machine that recognises damage, decides how to respond, physically restores itself and confirms that function has returned would need all of them working together.

Organisms make that integration look easy. Engineering is still building it piece by piece.

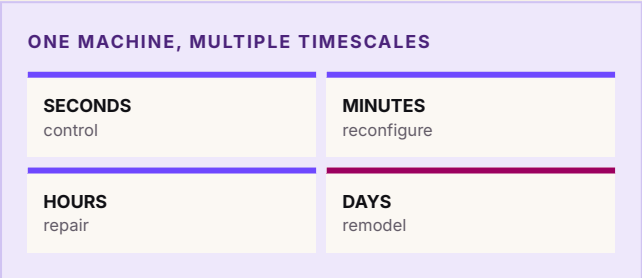
Living systems add another timescale

Article 3 covered biohybrid robots and engineered living systems. They belong here too, because living materials add development to the mix. Engineered muscle matures. Mechanical training changes its performance.

Cells reorganise, tissues remodel, and some living constructs show limited repair. Their state today depends partly on yesterday.

A modular robot changes because a controller commands it. A learning robot changes its behaviour with experience. A living component changes because biological development continues after assembly.

Future systems may combine all of these timescales: control adapting in seconds, morphology reconfiguring in minutes, materials repairing over hours, living tissue remodelling over days. Design stops being a single event and becomes a process running at different speeds throughout the machine.



And then there is evolution

The word needs care. Evolutionary robotics is already a substantial field. Researchers use algorithms inspired by variation, inheritance and selection to explore robot controllers, bodies or both. There have even been experiments where physical robot morphologies were automatically fabricated, tested, selected and modified across generations.^[8]

That is evolutionary design. It doesn't mean deployed robots currently reproduce, mutate and evolve on their own in open environments. Kinematic replication under laboratory conditions, as in the Xenobot work, is not open-ended artificial evolution.

The real picture is more interesting without the exaggeration. Automated morphology search exists. Physical testing feeds back into design. Controllers learn during deployment. Robots update models of their own bodies. Structures reconfigure. Distributed systems recognise damage. Materials begin repairing it. Living components develop. Those are separate capabilities today, and their combination would pro-

duce something quite different from a conventional robot long before autonomous artificial evolution becomes an issue.

Design time and run time begin to overlap

Software engineering has long separated what happens before execution from what happens while a program runs. Robotics inherited the split: at design time engineers decide what the robot is, and at run time the robot uses that design.

Physical AI is weakening that boundary. Look at the loops we now have. FAST puts AI in the design loop. Visual self-modelling lets a body inform its own control. FLAIR turns unexpected physics into adapted behaviour. Reconfigurable soft-robot control lets a new morphology bring a new policy. Cellular bricks let local modules recognise a collective shape and its damage.

The robot increasingly takes part in work that used to belong to its designers. That doesn't remove the engineer. It moves where engineering happens, and some of it moves into the deployed system.

The body becomes something intelligence can work on

This series began with a simple idea: Physical AI gives intelligence a body. Following that body through the research led somewhere more complicated. Biology can inspire its architecture. Morphology can carry part of its functionality. Living tissue can become one of its materials. Machines can connect directly to biological sensorimotor systems. And now the body itself can become dynamic: modelled, optimised, reconfigured, adapted around, monitored from within, partly repaired and developed over time.

No existing robot combines all of these. The research is still fragmented across materials, scales, laboratories and applications, and many demonstrations depend on controlled environments, external infrastructure or narrow tasks.

But the direction is visible. We have spent most of robotics designing intelligence for a body. Physical AI increasingly lets intelligence take part in designing, understanding and adapting the body itself. The old sequence stretches into a loop:

THE ROBOT IS NEVER QUITE FINISHED

DESIGN

DEPLOY

OBSERVE

ADAPT

RECONFIGURE

CONTINUE DESIGNING

A future physical intelligence may never have one final form. The robot may never be completely finished.

What to watch

These threads don't move at the same speed. Adaptive control, automated design loops and shared human-machine control are already being tested on real hardware. Reconfigurable and self-healing bodies are earlier. Living, self-maintaining machines are still a laboratory frontier.

The questions that come with them won't wait for the engineering. Who is accountable when an AI shares control of a prosthetic hand? Who owns the neural data flowing through it? What do we owe living components, and the animals used as platforms? Answers need to develop at the pace of the technology, which is not how such answers usually arrive.

Is the body of your robot still a decision you make once?

That is the end of this series. I would like to know which of these threads you think will reach real products first.

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